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Element recognition of laser-induced breakdown spectroscopy by comparing vectors of peak quantities

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ABSTRACT

Element recognition of laser-induced breakdown spectroscopy (LIBS) can be achieved by analyzing the corresponding peaks of plasma emissions. However, it is still challenging to make quick judgments due to database disagreements, such as wavelength or line-ratio. This study presents a simple method to conduct element recognition based on comparing vectors established by the quantities of spectral peaks, which allows an accurate recognition of the principle elements regardless of wavelength shifts. By examining deep-sea samples, this method demonstrates the recognition effectiveness for rare earth elements (REEs) with an accuracy over 80%. The developed method proves to be practical in LIBS qualitative analysis, and it can serve as an effective complement to existing approaches for element identification.

1. Introduction

Laser-induced breakdown spectroscopy (LIBS) is a versatile technique used for qualitative and quantitative analysis by referencing the elemental emissions during laser ablation of targets [1]. LIBS has applications in various fields including metallurgy [2,3], biomedicine [4–6], geology [7–10], forensic studies [11,12], and archaeology [13,14]. In these applications, the main use of LIBS is qualitative analysis, which is achieved by recognizing elements through the detection of elemental emissions in the LIBS spectra [15–17]. Element recognition is commonly accomplished through 'wavelength matching', where the wavelengths of emission lines are compared with those in the standard database (*e.g.,* NIST) [20,23–29]. The accuracy of wavelength measurement is crucial for the success of element recognition, because any wavelength shift would lead to recognition failure. Moreover, it is common to assign multiple elements to a single line when using wavelength matching, since different element emissions may occur at similar or identical wavelengths due to the resolution limitation of spectrometer. Therefore, there is a need to enhance the robustness of element recognition in LIBS qualitative analysis.

Labutin et al. and Yaroshchyk et al. proposed a solution as utilizing spectrum similarity rather than dwelling on the spectral wavelength [18,19]. The element recognition is achieved by assessing the similarity between the experimental spectra and the simulated spectra generated on the LIBS database (*i.e.,* NIST). However, prior knowledge of plasma temperature (T) and electron density (Ne) is required. Besides, Ukwatta et al. incorporated machine-learning techniques for element recognition, specifically as a classification application [21]. They utilized support vector machine (SVM) and artificial neural network (ANN) in this recognition process. A high accuracy of 99% was successfully achieved, but a deep optimization was necessary in the training phase.

Amato et al. introduced the concept of a 'text retrieval algorithm' in element recognition, with the 'vector space model' as its core [22]. In this algorithm, both the text document and the text query were considered as vectors consisting of weighted terms. The cosine of the angle between the query vector and the document vector was computed to determine the relevance. In the application of this algorithm to LIBS [22], the spectral peaks (wavelength, intensity) were treated as terms, while elements/samples were considered as documents/queries. The element recognition was achieved by ranking the results based on the

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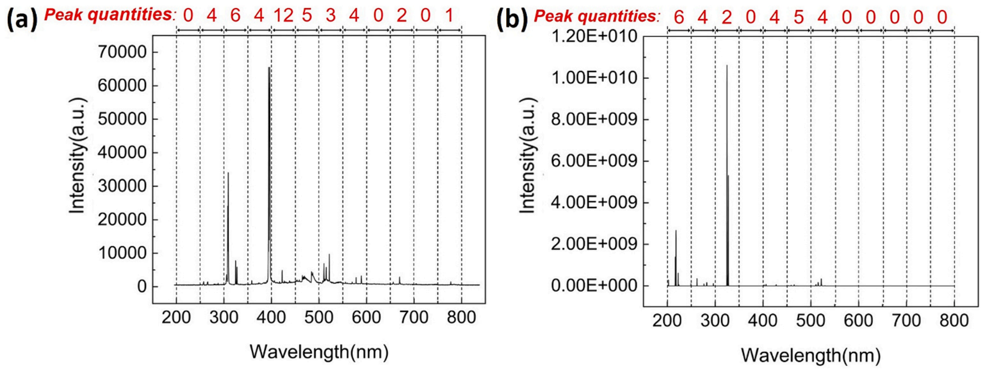

Fig. 1. LIBS spectrum after peak counting: (a) Cu—Al alloys; (b) NIST data of Cu.

calculation of the cosine angle between the sample vector and the elements vector. However, before executing the ranking, an appropriate weight value needed to be assigned to the sample vector. The success of the vector space model lies in its ability to convert the target relevance into a distance in a space coordinate, with the concept of a 'vector' being crucial in this process.

Spectral lines are commonly employed to identify atoms and molecules, the fundamental constituents of materials. Each spectral line corresponds to a transition between energy levels. Therefore, the number of spectral lines or peaks can indicate the number of energy levels, providing valuable information about atoms, molecules, or materials. Hence, in this work, we utilized the vector space model to recognize elements as well, and the vector was established based on quantities of spectral peaks.

2. Method principles

In the 'vector space model', it is critical to establish vectors. Differed from the publication [22], we propose to use peak quantities as the fundamental information for vector establishment. To accomplish this, the entire spectrum was divided into several spectral bands, and the peak quantities under each band were utilized as the vector element. As a result, a spectrum can be represented by a single vector comprising multiple element units, as formula (1).

$$\vec{V} = [x_1, x_2, x_3 \cdots x_n] \tag{1}$$

Where, x_i is the quantities of recognized peaks within the spectral band. To minimize the occurrence of "0" valued element, a variable band "d" is applied as below:

$$\begin{cases} 0 \le x_i \le 2, d = 50\ nm \\ 2 < x_i \le 5, d = 10\ nm \\ 5 < x_i < \infty, d = 5\ nm \end{cases} \tag{2}$$

In this manner, the value of vector element can be maintained in a reasonable range, ensuring comparability of the vector under the 'vector space model'. But it is noticed that the bandwidth variability is directly determined by the target spectrum. Based on that, the sample detection result is derived as a 'sample vector' $\vec{V_{sp}}$, and the NIST data of the specific element is similarly established as a 'standard vector' $\vec{V_{st}}$ under the same bandwidths of $\vec{V_{sp}}$. With these vectors, the relevance is determined through cosine calculation between $\vec{V_{st}}$ and $\vec{V_{sp}}$:

$$cos\left(\vec{V_{st}}, \vec{V_{sp}}\right) = \frac{\left\langle \vec{V_{st}}, \vec{V_{sp}} \right\rangle}{\left|\vec{V_{st}}\right|\left|\vec{V_{sp}}\right|} \tag{3}$$

In the given formula, $\left\langle \vec{V_{st}}, \vec{V_{sp}} \right\rangle$ represents the dot product of two vectors, and $\left|\vec{V_{sx}}\right|$ denotes the vector norm. When the cosine value is 1, it indicates that the sample vector $\vec{V_{sp}}$ coincides with the standard vector $\vec{V_{st}}$. This suggests that the sample emission can be attributed to the specific element under the standard vector $\vec{V_{st}}$. Conversely, when the cosine value is 0, there is no correlation between the sample vector $\vec{V_{sp}}$ and the standard vector $\vec{V_{st}}$. It is implied that there is no $\vec{V_{st}}$ element in the sample detection. Therefore, the cosine value would represent the possibility being recognized as the $\vec{V_{st}}$ element. In this work, the cosine value was used as the score for assessing the recognition result.

3. Results and discussion

Fig. 1 shows a typical LIBS spectrum of metal alloy (Al/Cu = 90/10) after peak counting. In this work, all detection results were obtained with a single channel spectrometer (AVANTES, AVASpec-UL2048)

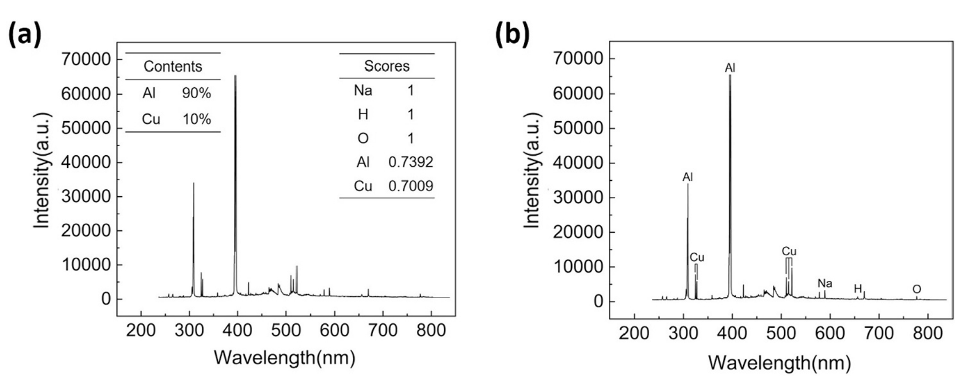

Fig. 2. Element recognition for Cu—Al alloys by using (a) proposed method and (b) Specline.

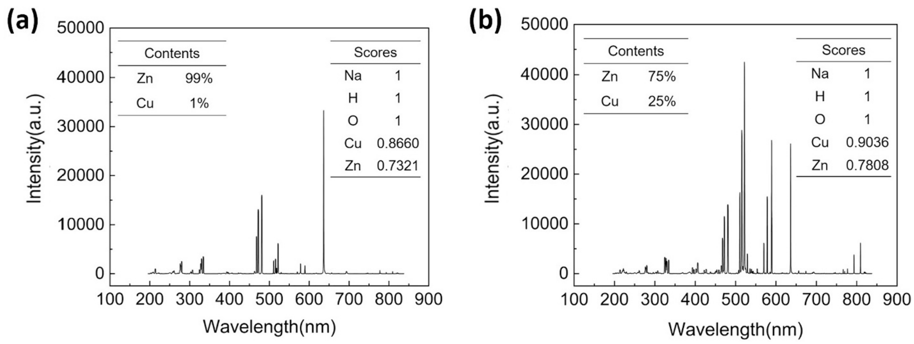

Fig. 3. Element recognition for Cu—Zn alloys: (a) Cu/Zn = 1/99; (b) Cu/Zn = 25/75.

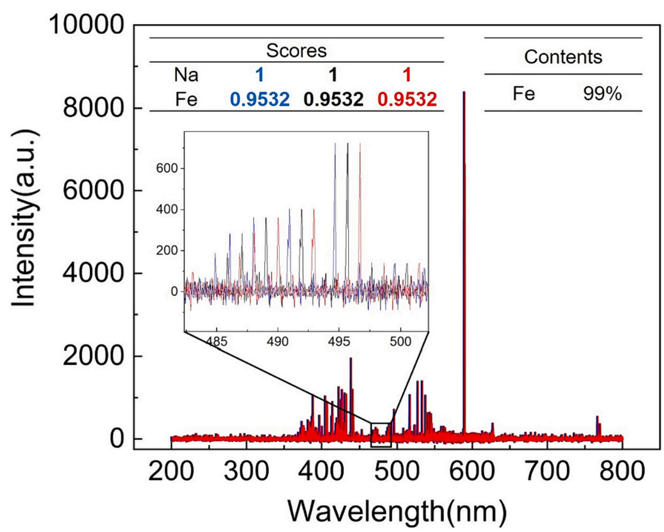

Fig. 4. Data in strikethrough have been retained. Please check if this is appropriate. Element recognitions under different wavelength-shift (Red: +1 nm shift; Blue: −1 nm shift). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

under the limited spectral resolution of 0.3 nm. In Fig. 1, the entire spectrum was divided into 12 spectral bands (50 nm / band) from 200 nm to 800 nm. After the peak counting in each band [30], illustrated in Fig. 1a, the spectrum was converted as the sample vector $\vec{V_{sp}} = [0,4,6,4,12,5,3,4,0,2,0,1]$. Under the same bands, the standard vector $\vec{V_{st}}$ of Cu element could be also established with the NIST data as $\vec{V_{st}} = [6,4,2,0,4,5,4,0,0,0,0,0]$, illustrated in Fig. 1b.

To improve comparability with the copper $\vec{V_{st}}$, the sample vector was simplified as $\vec{V}_{sp}{}' = [0,4,6,0,12,5,3,0,0,0,0,0]$, and the relevance calculation was focused on the $\vec{V_{st}}$ dimensions with non-zero value. As a result, the cosine value between the sample vector and the copper $\vec{V_{st}}$ was calculated as the recognition score of 0.7009. It is suggested in Fig. 2a that the copper content was probably existed in the metal alloy. Under the same procedure, the aluminum element was also confirmed as the alloys composition with a high score of 0.7392. As a comparison, we also employed the commercial software "Specline" to realize the element recognition, and the Specline was developed by PLASUS company using wavelength matching. The corresponding result is shown in Fig. 2b, and the element recognition was obtained when the key parameter "Hit interval" was set as 0.1 nm. The "Hit interval" refers to the wavelength difference between the target line and the database line in the wavelength matching.

Fig. 3 presents the element recognition for Cu—Zn alloys, including the recognition scores. The copper element was finely recognized with score of 0.8660, even the concentration was as low as 1%. In addition, the major element of zinc was successfully identified with the score of 0.7321. Despite of a significant change in elemental concentration, the recognition score was still preserved as shown in Fig. 2b. Furthermore, sodium (Na) was firmly recognized with hydrogen (H) and oxygen (O), and the proposed method can work for non-principle elements as well.

In Fig. 4, element recognition of the iron sample was carried out under the wavelength shift of 1 nm. It shows that our method was insensitive to this wavelength variation, even when dealing with the intricate emissions of iron. The recognition was finely achieved no matter the wavelength was increased or decreased. The recognition score stayed the same value even after the 1 nm drift. The proposed method was not built on the wavelength matching, so it is reasonable to achieve accurate element recognition in the case of wavelength shift. This merit might be useful in the field measurements, because the LIBS spectrometer is occasionally wavelength-drifted in the harsh environment.

To evaluate the applicability for complex targets in the real application, we used the deep-sea sediments (Table 1) to examine our method for the recognition of rare earth elements (REEs). These sediments are

Table 1
REEs contents in the deep-sea sediment samples.

Sample (GBW)	REEs contents / ppm															
	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Sc	Y
07586	32.9 ±1.5	74.5 ±2.3	9 ±0.3	38.4 ±1.8	8.7 ±0.4	2.14 ±0.07	8.8 ±0.3	1.46 ±0.06	8.2 ±0.4	1.67 ±0.07	4.83 ±0.23	0.72 ±0.04	4.6 ±0.25	0.73 ±0.05	14.7 ±0.8	48 ±3
07587	84.6 ±1.8	163 ±6	23.7 ±0.8	101 ±4	23 ±2	5.58 ±0.12	23.4 ±1.2	3.81 ±0.09	22.2 ±0.7	4.5 ±0.2	12.3 ±0.4	1.77 ±0.12	11.1 ±0.6	1.77 ±0.11	20.8 ±1	127 ±6
07588	113 ±5	217 ±10	31.9 ±1.2	138 ±5	31.6 ±1.2	7.82 ±0.37	31.6 ±1.1	5.26 ±0.17	30.6 ±0.9	6.2 ±0.4	16.9 ±0.6	2.42 ±0.14	15 ±1	2.37 ±0.16	25.5 ±1.4	178 ±5
07589	247 ±11	113 ±6	54.9 ±2.6	245 ±7	51.5 ±1.6	13.4 ±0.6	60.3 ±2.2	10.2 ±0.4	66.5 ±2.1	15.3 ±0.5	44 ±1.3	6.45 ±0.29	40.9 ±1.9	6.7 ±0.4	25.1 ±1.4	532 ±21
07590	348 ±12	132 ±4	76 ±4	342 ±13	71 ±3	18.9 ±0.8	85 ±4	14.3 ±0.6	92.8 ±2.8	21.4 ±0.8	62.3 ±2.1	9 ±0.5	56.8 ±2.6	9 ±0.4	32.2 ±1.4	732 ±29

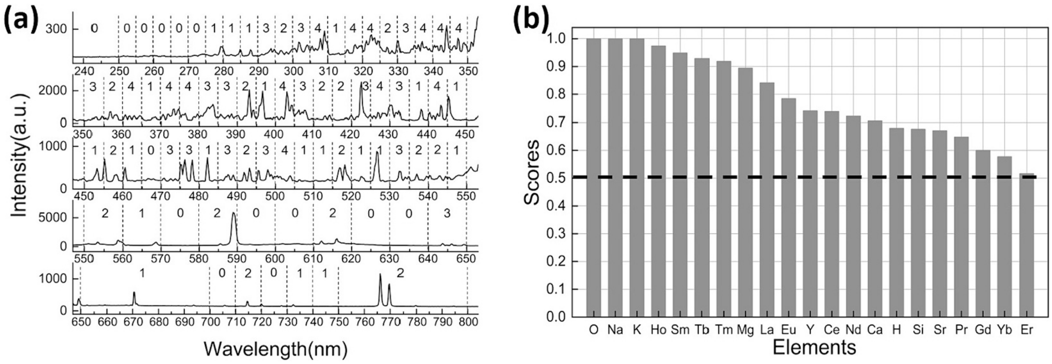

Fig. 5. Typical results under the proposed method: (a) Peak counting under variable bands; (b) Element recognition with the score above 0.5.

Table 2
REEs recognition using the proposed method.

Samples	Recognition ways	Recognized REEs	Unrecognized REEs	Recognition rate
GBW07586	Our method	Y, La, Pr, Ce, Nd, Sm, Eu, Gd, Tb, Ho, Tm, Yb, Lu	Er, Dy, Sc	81.25%
	Specline	Y, La, Pr, Ce, Nd, Sm, Eu, Gd, Tb, Ho, Tm, Yb	Er, Dy, Lu, Sc	75.00%
GBW07587	Our method	Y, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Ho, Er, Tm, Yb	Dy, Lu, Sc	81.25%
	Specline	Y, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Ho, Er, Tm,	Yb, Dy, Lu, Sc	75.00%
GBW07588	Our method	Y, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb	Lu, Sc	87.50%
	Specline	Y, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Ho, Er, Tm, Yb	Dy, Lu, Sc	81.25%
GBW07589	Our method	Y, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb	Sc	93.75%
	Specline	Y, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Ho, Er, Tm, Yb	Dy, Lu, Sc	81.25%
GBW07590	Our method	Y, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Ho, Er, Tm, Yb	Dy, Lu, Sc	81.25%
	Specline	Y, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Ho, Er, Tm, Yb	Dy, Lu, Sc	81.25%

standard samples (GBW 07586 ~ GBW 07590) with certified concentrations for 16 REEs (Y, La, Ce, Sc, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu). The sediment samples are offered by the First Institute of Oceanography (FIO) of China, and the elemental quantity uncertainty is varied from 0.04 ppm (GBW 07586, Tm) to 29 ppm (GBW 07590, Y). However, the multi-element in sediments would be a big challenge for the proposed way, because the low spectral resolution (0.3 nm) might cause more issues with complicated emissions.

Before using our method for element recognition, we simplified the LIBS database to include the REEs, the air components (C, H, O, N) and the seawater elements (O, H, Cl, Na, Mg, S, Ca, K, Br, Ca, Sr, B, F, Si). For

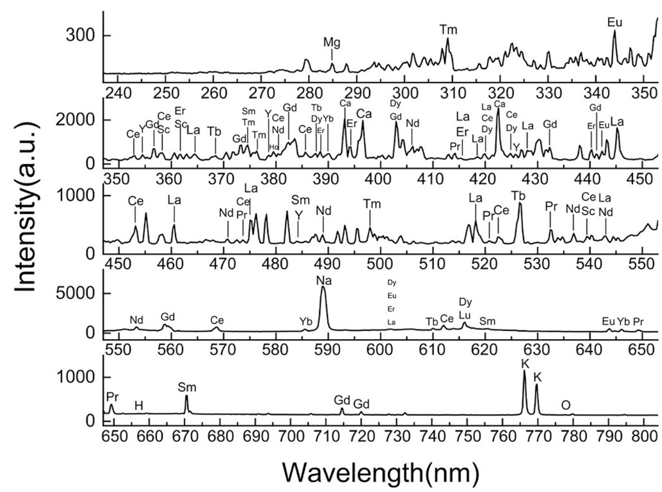

Fig. 6. Typical recognition result for REEs elements in the deep-sea sediment sample (GBW 07590) by using Specline software.

each REE sample, the sample vector $\vec{V_{sp}}$ was obtained under the variable bands (Fig. 5a), and the standard vectors $\vec{V_{st}}$ were established accordingly. Based on that, the sample vector $\vec{V_{sp}}$ was sequentially computed the score with a series of standard vectors $\vec{V_{st}}$.

Fig. 5b is the recognition score after using the threshold of 0.5. It is shown that REEs were successfully recognized as: Y, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Ho, Er, Tm, Yb, Lu. But, the elements (Dy, Lu, Sc) were not recognized. Besides that, the seawater elements (O, Na, K, Mg, Ca, H, Si) were found in the detection result. It indicates that the proposed method can be an alternative way to achieve the element recognition. Table 2 is the recognition result for 5 sediment samples of REEs. Over 80% REEs were successfully recognized, and highest recognition rate was up to 93.75%. However, some elements (Sc, Lu, Dy) were difficult to be recognized due to their poor emission intensity for detection.

In comparison, the Specline software was also utilized for REEs recognition with the optimized "Hit interval" value as 0.1 nm. However, as shown in Table 2, the recognition rate was lower compared to the results obtained using the proposed method. This discrepancy arises from the fact that multiple elements are often assigned to a single spectral peak when employing Specline, making it challenging to determine accurate element recognition (Fig. 6). Furthermore, increasing the "Hit interval" to higher values exacerbates elemental overlapping.

4. Conclusion

This work showed that our method could offer a general recognition for elements without considering the peak overlap or the wavelength shift. Element recognition is a common topic in LIBS, and there are different ways to achieve it. In this work, we develop an approach to realize the element recognition *via* comparing the vector relevance, while the vector was established on peak quantities. In this way, the element recognition could be easily implemented by referencing the cosine value between the sample vector $\vec{V_{sp}}$ and the standard vector $\vec{V_{st}}$. It is found that the proposed way is immune to the wavelength shift. For the multi-composition analysis, our method was capable to recognize most elements in the marine samples, and the REEs recognition could reach the highest rate of 93.75%. However, it is noticed that the proposed method is an approach to determine the element contents rather than a way to identify elements with specific wavelengths. The future prospects of our method lie in its potential to narrow down the search scope of elements, resulting in a reduction of overlapping assignments during element identification when combined with 'wavelength matching' approaches. It should be noted that 'wavelength matching' remains a direct means of identifying elements (*e.g.*, Specline), while the proposed method offers a more general approach to element recognition.

CRediT authorship contribution statement

Xuanbo Zhang: Writing – original draft, Methodology, Investigation, Formal analysis. **Shoujie Li:** Validation, Methodology, Investigation, Formal analysis. **Zengfeng Du:** Resources. **Wangquan Ye:** Methodology. **Jinjia Guo:** Funding acquisition. **Ye Tian:** Methodology. **Xueshi Bai:** Writing – review & editing, Validation. **Vincent Detalle:** Validation. **Ronger Zheng:** Supervision. **Xin Zhang:** Project administration. **Yuan Lu:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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